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BE (3rd Semester)

Examination, April-May, 2013

Branch : CSE, IT, Mechatronics

DIGITAL ELECTRONICS & LOGIC DESIGN

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : All questions carry equal marks. Part 'a' in each question is compulsory. Attempt any two parts from 'b', 'c' and 'd'.

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UNIT - I

Q. 1. (a) What is meant by a weighted binary code. Explain using weighted codes of 5421 system for decimal digits 0 to 9. 2

(b) Use 2's complement and subtract $11101_2 - 1101_2$. 7

(c) Add the following using excess 3 addition : 7

(i) $8_{10} + 6_{10}$

(ii) $81_{10} + 36_{10}$

(d) Minimize using K-Map and implement using logic gates (AOI logic) 7

$$f = \sum_{m} (0, 2, 3, 8, 10, 12) + d(4, 6, 7)$$

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UNIT - II

Q. 2. (a) Define multiplexer, what is the difference between data inputs and select inputs of a multiplexer. 2

(b) Use a 8 : 1 multiplexer to implement the logic for the function. 7

$$f = \sum m(0, 1, 2, 3, 7, 9, 10, 12, 13, 15)$$

(c) Design a BCD adder that adds two 4 bit BCD numbers. 7

(d) Design a 4 bit binary to gray code converter. 7

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UNIT - III

Q. 3. (a) What is meant by Fan in and Fan out of logic gate. 2

(b) Describe various characteristics that are used to evaluate various digital logic families. 7

(c) Draw a two input TTL NAND gate and explain its operation. 7

(d) Write short notes on ECL. 7

UNIT - IV

Q. 4. (a) Write comparison between combinational circuits and sequential circuits. 2

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(b) Convert a JK flip flop to T flip-flop. 7

(c) Explain triggering of flip flop. 7

(d) Design a BCD counter, asynchronous in nature. 7

UNIT - V

Q. 5. (a) What is a dynamic RAM ? 2

(b) Write short notes on Successive Approximation Analog to Digital Converter. 7

(c) Explain the operation of a R-2R ladder digital to analog converter. 7

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(d) A combinational circuit is defined by functions 7

$F_1 = \sum m(1, 5, 7)$ and $F_2 = \sum m(5, 6, 7)$

Implement the circuit with a PLA.

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